SMARTSTEP: A SMART PIEZOELECTRIC ENERGY HARVESTING FLOOR WITH IOT-BASED POWER MONITORING AND BATTERY MANAGEMENT SYSTEM

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

RATIONALE OF THE STUDY

In the modern global landscape, the rapid depletion of conventional energy resources and the increasing levels of greenhouse gas emissions have intensified the global energy crisis (Wang & Azam, 2024). As a result, there has been a growing emphasis on the transition toward renewable and sustainable energy sources. While large-scale renewable infrastructures such as solar, wind, and hydroelectric systems contribute significantly to energy generation, their implementation often involves high installation costs and efficiency challenges, particularly in energy storage and distribution (Lozano et al., 2019; Supapo et al., 2024). Consequently, there is a growing interest in decentralized energy harvesting technologies that can generate power from readily available environmental sources such as mechanical vibrations and human activity (Pradeesh et al., 2022).

In the Philippine context, the electricity demand continues to increase, with national consumption projected to reach approximately 163,639.9 GWh by 2030 (Parreño, 2022). This demand is especially critical in off-grid and rapidly urbanizing areas where reliable power supply remains limited (Delima, 2019).

Existing renewable systems in the country also face efficiency challenges. For example, studies on solar energy systems in Cebu have reported significant energy losses when storage systems reach full capacity, resulting in wasted energy and reduced system effectiveness (Supapo et al., 2024). These limitations highlight the need for alternative, localized energy solutions that can be integrated directly into existing infrastructure.

At the local level, Cebu presents both challenges and opportunities in energy utilization. In off-grid communities such as Gilutongan Island, reliance on diesel generators has resulted in high operational costs and limited access to electricity (Lozano et al., 2019). Meanwhile, urban areas experience high pedestrian traffic, where mechanical energy generated from human footsteps remains largely untapped (Sharma et al., 2022). The direct piezoelectric effect provides a potential solution by converting mechanical stress into electrical energy, offering a viable method for harvesting energy from everyday human activities (Covaci & Gontean, 2020).

Despite these developments, existing renewable energy systems often fail to efficiently capture and utilize small-scale, decentralized energy sources such as human kinetic activity. Furthermore, current systems lack integrated monitoring and intelligent energy management mechanisms, resulting in energy wastage, inefficient storage, and limited practical application in real-world environments.

This gap highlights the need for a system that not only harvests energy efficiently but also manages and monitors it intelligently.

In response to these challenges, this study aims to design and develop SmartStep, a smart piezoelectric energy harvesting floor integrated with an Internet of Things (IoT)-based power monitoring and battery management system. The proposed system will convert mechanical energy from human footsteps into electrical energy, regulate and store the generated power using an intelligent battery management system, and provide real-time monitoring through an IoT-enabled platform.

STATEMENT OF THE OBJECTIVES

General Objectives:

The primary objective of this study is to design, develop, and evaluate SmartStep, an IoT-enabled piezoelectric energy harvesting system, in terms of energy efficiency, functionality, and system performance.

Specific Objectives:

-Design a piezoelectric-based system capable of converting mechanical footstep pressure into electrical energy;

-Develop an intelligent battery management system to regulate energy storage and prevent overcharging or energy loss;

-Implement an IoT-based monitoring system for real-time tracking of energy generation and system performance;

-Develop a mobile dashboard to visualize energy data, including generated power and battery status;

-Evaluate the system's efficiency by comparing theoretical and actual energy output.

SIGNIFICANCE OF THE STUDY

The significance of SmartStep lies in its ability to transform passive urban infrastructure into an active, data-generating energy asset and a functional power utility. By converting kinetic energy from footsteps into regulated electricity, this study demonstrates that human activity is a viable, carbon-neutral component of the modern green energy mix. This research is of particular importance to the following beneficiaries:

- **End Users (Pedestrians and Urban Residents)** – Residents of high-traffic urban areas benefit from the integration of sustainable technology that does not disrupt their daily routines. By providing a low-cost, supplementary power source for small-scale electronics and emergency sensors in public spaces, the system enhances convenience and accessibility to "green" energy without the high capital costs associated with traditional solar or wind installations (Selim et al., 2024).
- **Local Organizations and Institutions** – Public and private institutions in Cebu can utilize SmartStep as a supplementary power hub to reduce dependence on the main power grid and lower operational costs. By implementing the Intelligent Battery

Management System (BMS), organizations can overcome the 24.45% energy loss typically seen in off-grid renewable installations, ensuring that every harvested watt is stored and made available for user charging rather than being wasted during "battery full" conditions.

- **Administrators and Decision-Makers** – Administrators benefit from the system's advanced data analytics, which track energy generation alongside charging station usage and battery health. This transparency allows for better-informed decision-making regarding maintenance and energy distribution. Automated reporting helps decision-makers optimize urban energy grids and evaluate the practical demand for localized charging points based on pedestrian density.
- **Researchers and Students** – This study serves as a technical reference for future research in the fields of IoT-integrated renewable energy and smart charging infrastructure. By successfully implementing "Deep Sleep" modes to reduce standby consumption by 80% and managing erratic AC spikes through smart circuitry to provide a stable charging output, this project provides a blueprint for academic inquiry into energy-neutral smart city systems.

SCOPE AND LIMITATION

Scope and Limitations:

The study will focus on the design and development of a SmartStep prototype integrated with a battery management system and IoT monitoring platform. It includes the development of a corresponding Intelligent Battery Management System (BMS) to regulate and store harvested energy. The research focuses on electrical generation via piezoelectric sensors and providing real-time data analytics through a dedicated mobile monitoring application. The system is designed to provide a regulated power output suitable for charging low-power IoT devices and small electronics.

Despite its innovative design, the SmartStep system is defined by several technical constraints. Its real-time monitoring is strictly network-dependent, requiring a stable Wi-Fi environment for the ESP8266 to prevent data gaps on the dashboard. Processing constraints also necessitate external regulation to protect the microcontroller's single analog input from high-voltage piezoelectric spikes to ensure system longevity, the researchers will integrate a Zener diode clamping circuit. This hardware-level protection shunts transient voltage spikes from the

PZT sensors that exceed the 3.3V/5V logic threshold, protecting the ESP8266 and INA219 from catastrophic failure. Regarding utility, the system's charging capacity is limited to low-current applications, making it unsuitable for high-wattage devices like laptops.

Furthermore, the factor of self-consumption means the system is most effective in high-traffic areas, where frequent harvesting events can offset the energy required to power the IoT and BMS components (Moussa et al., 2022; Chen et al., 2021). Finally, long-term durability remains a factor, as continuous mechanical impact may eventually lead to material fatigue in the piezoelectric elements (Sharma et al., 2022).

CONCEPTUAL FRAMEWORK

Input–Process–Output (IPO) Model

The study follows an Input–Process–Output (IPO) model, where mechanical energy from footsteps is converted into electrical energy through

piezoelectric sensors, processed through energy conditioning and storage systems, and delivered as usable power and real-time monitoring outputs.

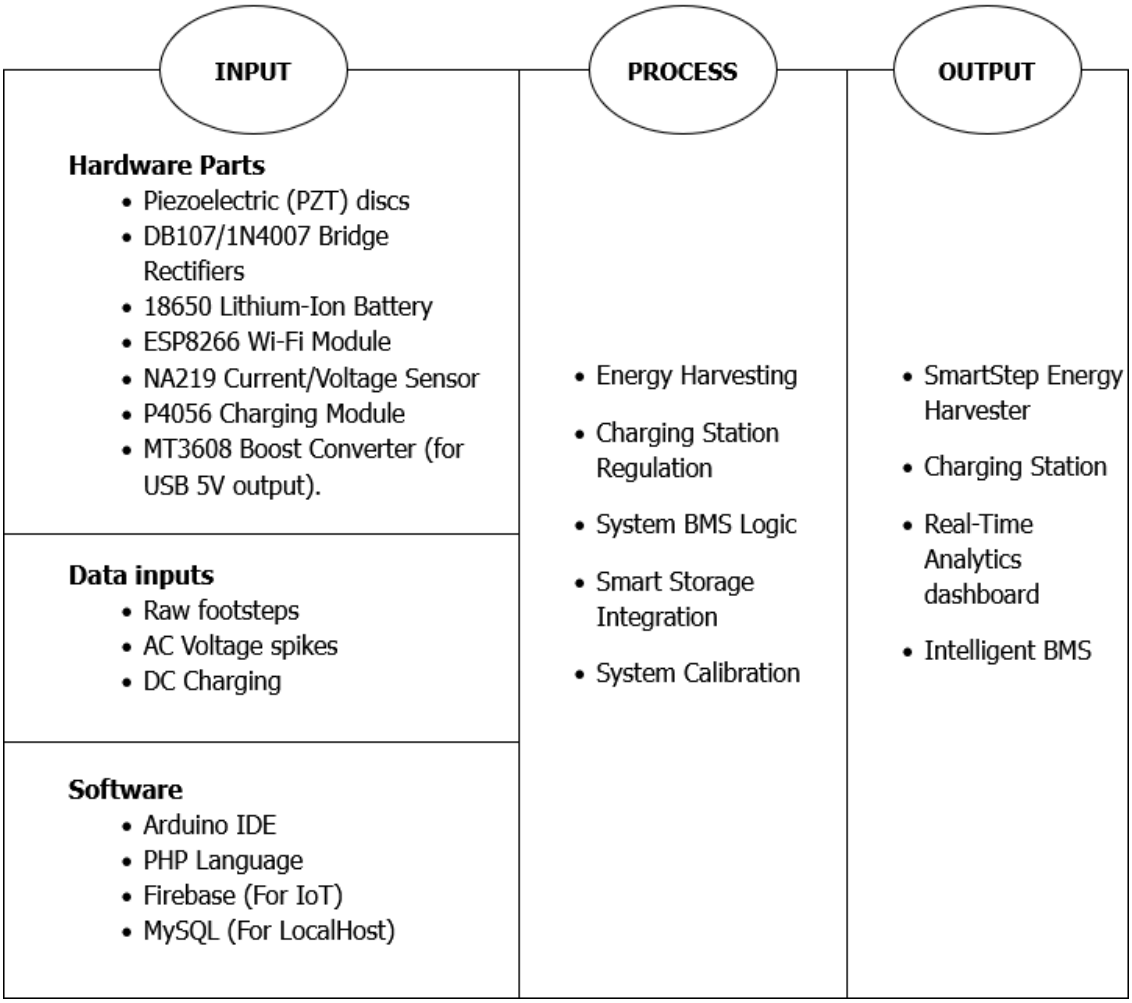

Fig 1. IPO Model

Input:

Input phase consists of the essential hardware and software requirements needed to build the system. This includes the Piezoelectric (PZT) array, the 18650 Lithium-Ion battery for storage, and the ESP8266

Wi-Fi module for connectivity. These are supported by software tools like the Arduino IDE and Firebase for data handling. The inputs are driven by a specialized knowledge base on the direct piezoelectric effect and Constant Current/Constant Voltage (CC/CV) charging profiles, ensuring that the raw data—such as mechanical pressure and voltage spikes—is correctly interpreted (Song et al., 2019; Jeon et al., 2020).

Process:

The process phase involves the technical transformation of these inputs into a functional system through five critical stages. First, energy harvesting and conditioning convert erratic AC spikes from the mat into stable DC power using bridge rectifiers. This is followed by Smart Storage Integration, where the TP4056 module manages the charging of the 18650 battery, and the INA219 sensor monitors the energy flow. The system then regulates this power through an MT3608 boost converter to provide a steady 5V USB output. Simultaneously, the software logic optimizes power consumption through deep sleep modes while reporting live analytics to the cloud, culminating in a system calibration to ensure stability under load.

Output:

Lastly the Output represents the tangible results of the study, resulting in a portable, modular SmartStep Energy Harvester. This hardware functions as an IoT-Enabled Charging Station, providing a self-sustaining power hub for low-power devices. Accompanying the hardware is an Intelligent BMS that safeguards battery health and prevents energy wastage. Finally, a Real-Time Analytics Dashboard provides a user-friendly mobile interface, displaying live metrics such as cumulative energy harvested, battery percentage, and pedestrian traffic counts.

In conclusion, this IPO-structured approach ensures that every step—from the physical impact of a footstep to the digital display on a smartphone—is technically sound and academically grounded. By bridging the gap between raw energy harvesting and user-centric IoT management, the SmartStep system provides a viable and transparent solution for sustainable urban power.

REVIEW OF RELATED LITERATURE AND STUDIES

RELATED LITERATURE

The continuous consumption of natural resources has forced a global prioritization of renewable energy transitions; however, many developing nations continue to face the challenge of rising electricity demand driven by rapid population growth and industrial expansion (Parreño, 2022). As traditional power grids struggle to keep pace, researchers are identifying decentralized methods to generate electrical energy, among which piezoelectric harvesting has emerged as a premier cost-effective solution for urban environments. Unlike large-scale solar or wind farms, piezoelectric systems can be embedded directly into high-traffic public infrastructure, such as floor tiles, to capture the kinetic energy of human footsteps that would otherwise dissipate as waste heat (Abadi et al., 2018; Moussa et al., 2022).

Recent advancements in material science, specifically regarding Lead Zirconate Titanate (PZT) ceramic sensors, have significantly improved the feasibility of this technology. Selim et al. (2024) demonstrate that modern PZT arrays can achieve energy conversion rates high enough to provide practical

power for micro-electronics at a fraction of the cost of older electromagnetic systems. Furthermore, the integration of these harvesters into Internet of Things (IoT) frameworks allows for the transition from passive energy collection to intelligent, data-driven systems. As noted by Nand et al. (2021) and Wang and Liang (2025), the ability to monitor energy yields in real-time through mobile applications and cloud dashboards is now essential for the sustainability of smart-campus initiatives. Beyond power generation, the resilience of these systems relies on sophisticated Intelligent Battery Management Systems (BMS). Modern literature emphasizes that a BMS is no longer just a peripheral component but a critical "responsive ecosystem" required to protect Lithium-Ion cells from the erratic input spikes typical of harvesting applications while ensuring long-term battery health (Anandha Kumar et al., 2021; Gabbar et al., 2021). Consequently, this segment discusses contemporary literature in the areas of Energy Harvesting, Battery Management, and IoT Analytics to serve as the technical and theoretical foundation for the development of SmartStep.

RELATED STUDIES

Several contemporary studies have explored the mechanical and electrical viability of harvesting energy from human locomotion. Research consistently demonstrates the high-voltage potential of Lead Zirconate Titanate (PZT) sensors when subjected to the mechanical strain of human footsteps. Abadi et al. (2018)

prototyped a 30 cm×30 cm tile containing 20 parallel-connected PZT transducers that successfully generated an AC voltage of up to 71.20 V. Their findings indicated an average power yield of 0.0604 watts per 10 footsteps, confirming that pedestrian kinetic energy is a viable, albeit intermittent, source for low-voltage loads. Similarly, Selim et al. (2024) designed an expanded 455 mm×405 mm tile that successfully extracted up to 249.6 milliwatts. These studies emphasize that while individual steps produce micro-pulses, strategic series and parallel sensor combinations are essential to upgrade the current and voltage levels for practical application.

Beyond energy collection, the integration of the Internet of Things (IoT) addresses the lack of operational transparency found in traditional energy harvesters. Nand et al. (2021) utilized a NodeMCU (ESP8266) as a central gateway to synchronize hardware parameters with an Android application. Their prototype facilitated the remote monitoring of real-time metrics, including voltage, current, and battery levels. This technical approach aligns with the "Smart City" framework, where infrastructure provides both renewable power and actionable data. To enhance this data-driven reliability, Krishna et al. (2024) demonstrated that utilizing specialized hardware gateways to log the State of Health (SoH) and cycle counts to cloud servers can significantly reduce system downtime. Furthermore, Qahtan et al. (2022) validated the use of platforms like

Blynk for real-time visualization, allowing users to monitor consumption currents and capacity directly from mobile devices.

Despite these advancements, a major challenge identified in recent literature remains the inherent instability of piezoelectric waveforms and the resulting risk to battery longevity. Selim et al. (2024) noted that a standard full-wave bridge rectifier is often insufficient for sensitive electronics, necessitating additional control circuits and specialized Schottky diodes to mitigate unstable output. To address hardware modularity and fault tolerance, Canilang et al. (2022) proposed a modular BMS topology that maintains high monitoring accuracy even when individual modules are at fault, achieving a DC-DC conversion efficiency of 92.74%. This modular approach is complemented by safety-focused studies like those of Bekti Firmanto et al. (2024), who integrated INA219 current sensors and Fuzzy Logic to monitor battery health with a capacity testing error of less than 1.8%. For storage, current research justifies a hybrid architectural approach: utilizing capacitors for immediate, instantaneous charge capture and Lithium-Ion batteries managed by specialized modules like the TP4056 for long-term storage and protection (Xie & Zhang, 2021).

RELATED SYSTEMS

The development of an effective SmartStep requires intense research on the current use of piezoelectric energy harvesting to draw useful learnings about functionalities that optimize energy harvesting and battery management functions. Through a consideration of well-known systems such as Pavegen and Waynergy, as well as academic prototypes, we are able to discern important learnings and best practices.

Pavegen serves as the primary commercial industry benchmark, utilizing electromagnetic and piezoelectric technology to generate between 1W and 10W per step. Its most significant feature is a sophisticated wireless API that allows for real-time analytics and data transmission, providing a blueprint for the "Smart" aspect of urban harvesters (Nand et al., 2021). However, its implementation is hindered by extremely high fabrication costs and a permanent, heavy infrastructure requirement that limits its use to large-scale urban projects (Li & Strezov, 2014).

Conversely, Waynergy offers a pavement-integrated solution primarily intended for industrial direct loads, such as street lighting and security sensors (Duarte et al., 2013). While its core functionality emphasizes extreme durability in outdoor urban environments, it lacks a user-centric charging interface and mobile transparency, prioritizing industrial-scale utility over personal device accessibility (Li & Strezov, 2014).

Meanwhile, academic prototypes by Nand et al. (2021) and Selim / Abadi (2018/2024) provide the foundation for low-cost implementation. Abadi et al. (2018) and Selim et al. (2024) successfully demonstrated that parallel-connected PZT arrays can generate significant AC voltages, reaching up to 71.20V, using affordable ceramic sensors. Furthermore, Nand et al. (2021) introduced the essential use of the NodeMCU (ESP8266) and the XL6009 boost converter to bridge the gap between unstable hardware output and consistent mobile application monitoring.

Table 1
Comparative Matrix For Related Systems

Features / Systems	**Pavegen**	**Waynergy**	**Nand et al.**	**Selim / Abadi**	**SmartStep (Proposed)**
1. Piezoelectric-Based	✗	✓	✓	✓	✓
2. Li-ion Battery Storage	✓	✓	✓	✗	✓
3. Regulated USB Output (5V)	✓	✓	✗	✗	✓
4. IoT Cloud Connectivity	✓	✗	✓	✗	✓
5. Mobile App Dashboard	✓	✗	✓	✗	✓
6. Intelligent BMS	✓	✗	✗	✗	✓

Logic					
7. Modular / Portable Mat	✗	✗	✗	✓	✓
8. Real-Time Energy Analytics	✓	✗	✓	✗	✓
9. Pedestrian Traffic Counter	✓	✗	✗	✗	✓
10. Low-Cost Fabrication	✗	✗	✓	✓	✓
Total Score (%)	**60%**	**30%**	**60%**	**40%**	**100%**

Legend:

✓ - Applicable

✗ - Non-Applicable

Synthesis of the Reviewed Literature and Studies

The reviewed literature and systems collectively suggest that piezoelectric energy harvesting is a mature enough technology to support small-scale electronic loads. There is a clear consensus among researchers that human footsteps in high-traffic areas represent a viable "waste" energy source that can be reclaimed through strategic sensor placement (Abadi et al., 2018; Moussa et al., 2022). Technical studies emphasize that while PZT sensors are cost-effective, their output is inherently erratic and high-voltage in nature,

necessitating robust power conditioning through rectifiers and boost converters to be useful for modern devices (Selim et al., 2024; Nand et al., 2021).

Recent research has significantly shifted the focus from mere energy extraction to the long-term sustainability of the storage medium. Modern literature highlights that a successful harvesting system must integrate an Intelligent Battery Management System (BMS) to bridge the gap between unstable energy inputs and sensitive Lithium-Ion storage (Gabbar et al., 2021). Breakthroughs in health monitoring, such as the use of Fuzzy Logic and specialized current sensors, have proven that real-time monitoring of State of Charge (SoC) and State of Health (SoH) can reduce capacity estimation errors to less than 1.8%, effectively extending battery lifespans (Bekti Firmanto et al., 2024; Anandha Kumar et al., 2021).

Furthermore, the shift toward "Smart Cities" has established the necessity of integrating IoT frameworks to provide users with transparency and real-time data analytics (Wang & Liang, 2025). The use of wireless protocols and cloud-based gateways now allows for a "responsive" energy environment where hardware performance—including terminal voltage and cycle counts—is monitored remotely to ensure reliability (Krishna et al., 2024; Qahtan et al., 2022).

When considering the ideal cost-effective solution, integrating the strengths of these systems is crucial. This approach must prioritize

comprehensive energy capture, a fault-tolerant modular architecture that ensures efficiency even under partial failure (Canilang et al., 2022), and user-centric data visualization through dedicated IoT dashboards.

RESEARCH GAPS

Despite the advancements in piezoelectric technology, several critical research gaps persist that SmartStep seeks to address. First, there is a significant socio-economic barrier where commercial solutions like Pavegen are prohibitively expensive for community-level or educational deployment, while low-cost academic prototypes often remain in a laboratory "bench-top" state without a rugged, deployable design.

Second, a critical management-protection gap exists; most existing low-cost prototypes focus strictly on basic data transmission but fail to integrate an Intelligent Battery Management System (BMS). Without a dedicated BMS, the 18650 Li-ion storage cells are vulnerable to the high-voltage spikes and erratic current characteristic of piezoelectric generation, leading to rapid battery degradation. Current literature suggests that while many systems monitor voltage, there is a significant lack of integrated logic for tracking the State of Health (SoH) and State of Charge (SoC) simultaneously to prevent deep discharge—a leading cause of Lithium-Ion failure in renewable applications (Anandha Kumar et al., 2021; Bekti Firmanto et al., 2024).

Third, there is a reliability and efficiency gap in power conditioning. While standard converters are common, most existing harvesting prototypes lack fault-tolerant modularity; if one sensor or battery module fails, the entire system typically goes offline. Recent industrial IoT research emphasizes the need for modular topologies that can maintain high conversion efficiency (exceeding 90%) even under partial fault conditions, a feature currently absent in portable energy mats (Canilang et al., 2022).

Finally, there is a lack of portability and operational transparency in current infrastructure-based systems, which are typically permanent installations requiring major construction. Existing studies often fail to provide a unified user-centric interface that visualizes both the energy harvested and the real-time safety status of the storage unit (Krishna et al., 2024; Qahtan et al., 2022). SmartStep addresses these gaps by synthesizing high-voltage PZT harvesting, a fault-tolerant modular architecture, an intelligent BMS with SoH monitoring, and a real-time IoT mobile dashboard into a single, portable modular mat solution.

RESEARCH METHODOLOGY

RESEARCH DESIGN

The researchers utilize the Agile Methodology for the development and implementation of SmartStep. The Agile approach is chosen for its flexibility and iterative improvement, enabling ongoing feedback and adjustment during the project lifecycle. This is particularly crucial for SmartStep, as it ensures that the physical hardware (PZT sensors and power circuitry) and the digital software (Blynk IoT dashboard) are synchronized effectively to meet the energy needs of users in high-traffic environments.

Agile is defined by incremental development, wherein the system is built in functional "sprints." Each cycle produces a working module—starting from raw energy capture and moving toward a fully regulated charging station—allowing for early detection of hardware inefficiencies or voltage instabilities. In addition, the development process remains flexible to cater to changes in hardware requirements or electrical priorities.

A. Power Acquisition and Energy Management

SmartStep's power acquisition entails the integration of a Lead Zirconate Titanate (PZT) sensor array with a specialized power conditioning circuit. This system converts mechanical strain into stabilized DC voltage, which is then managed by an Intelligent Battery Management System (BMS). To ensure system longevity, the researchers will integrate a Zener diode clamping circuit. This hardware-level protection shunts transient voltage spikes from the PZT sensors that exceed the 3.3V/5V logic threshold, protecting the ESP8266 and INA219 from catastrophic failure.

For effective data management, the project utilizes the ESP8266 microcontroller to log terminal voltage and current via the INA219 sensor. This provides secure, real-time tracking of the battery's State of Charge (SoC) and State of Health (SoH), ensuring that the stored energy is used efficiently and that the 18650 Li-ion cells are protected from over-discharge (Anandha Kumar et al., 2021).

B. System Development Phases

Fig 2. Agile Development Methodology

I. Sprint I: Requirement Planning

This initial phase establishes the technical pivot and design constraints for the SmartStep architecture, transitioning from conceptual goals to high-fidelity engineering specifications. The researchers define the electrical benchmarks by calculating the Electromechanical Coupling Factor (k33) of the 35mm PZT sensors, determining the minimum footfall frequency and mechanical force required to overcome the forward voltage

drop of the rectification stage and maintain a positive State of Charge (SoC).

The selection of 35mm Lead Zirconate Titanate (PZT) transducers is based on an analysis of their resonant frequency and capacitive reactance, ensuring the sensors are optimized for the low-frequency, high-impact nature of human gait. Significant focus is placed on the impedance matching between the high-impedance PZT array and the low-impedance battery storage to minimize energy reflection and maximize power transfer.

Critical planning is further devoted to the Intelligent Battery Management System (BMS) logic. The researchers establish a rigorous safety profile by defining the upper and lower threshold voltages (Vmax and Vmin) for the 18650 Lithium-ion cells. These thresholds are calibrated to prevent dendrite formation during deep discharge and thermal runaway during over-voltage spikes.

Furthermore, the planning phase specifies the Coulomb counting algorithms or Open Circuit Voltage (OCV) look-up tables that the ESP8266 will utilize to estimate battery longevity. This sprint concludes with a finalized Bill of Materials (BOM) and a prioritized Firmware Logic Flow,

detailing the interrupt-driven sampling rate for the INA219 sensor and the asynchronous data transmission intervals for the IoT gateway.

II. Sprint II: Design and Prototyping

In the second sprint, the focus shifts to the physical and electrical topology of the SmartStep system. The researchers execute a spatial layout of the PZT transducers within a modular grid, performing a calculation of the impedance mismatch to determine the optimal ratio of series and parallel connections. This configuration is critical to balance the Open-Circuit Voltage (Voc)—which can reach high amplitudes—and the Short-Circuit Current (Isc) to ensure a compatible input for the downstream power electronics.

The electrical design incorporates a DB107 Schottky-type Bridge Rectifier and a 50V 1000μF Low-ESR (Equivalent Series Resistance) Electrolytic Capacitor. This combination functions as a passive smoothing filter (C-filter), designed to mitigate voltage ripple and rectify the stochastic, high-frequency AC transients into a filtered DC signal. The choice of a 50V rating is a deliberate safety measure to prevent dielectric

breakdown within the capacitor during peak impulse events, such as a high-impact footstep.

Simultaneously, a CAD-based structural model of the modular mat is developed, focusing on load distribution and stress analysis. The design must satisfy the dual-constraint of low-profile portability and mechanical resilience. The housing is engineered to protect the sensitive INA219 current-shunt monitor and the interconnecting busbars from mechanical shear and compressive fatigue. By simulating the mechanical-to-electrical transduction path, the researchers ensure that the displacement of the tile is sufficient to trigger the PZT's piezoelectric effect without exceeding the material's yield strength, thereby preventing micro-fractures in the ceramic elements.

III. Sprint III: Development (Hardware-Software Integration)

This stage represents the operationalization of the system’s logic, where the hardware "Process" and software firmware are merged. The researchers assemble the power rail, integrating the TP4056 linear charger to manage the Constant-Current/Constant-Voltage (CC/CV) charging profile of the 18650 Li-ion battery. To ensure the stored potential is accessible for consumer electronics, an MT3608 High-Efficiency Boost

Converter is implemented and tuned via its potentiometer to maintain a regulated 5V DC output regardless of the battery's fluctuating terminal voltage.

Simultaneously, the firmware for the ESP8266 (NodeMCU) is developed using an asynchronous programming model. The microcontroller interfaces with the INA219 current-shunt monitor via the I2C (Inter-Integrated Circuit) protocol, sampling high-side voltage and bus current at a high frequency. This data is then processed to calculate the State of Charge (SoC) using a combined Coulomb Counting and Voltage-Lookup algorithm. The sprint concludes once the ESP8266 successfully establishes a TCP/IP handshake with the cloud gateway, confirming that real-time telemetry is ready for visualization.

IV. Sprint IV: Deployment and User Testing

The SmartStep prototype will be deployed at the faculty entrance, a high-traffic strategic hub characterized by high-density, repetitive footfall. This environment allows the researchers to evaluate the system's performance under intermittent stochastic loading. The modular mat will be integrated into the existing walkway, while the charging station and IoT monitoring hub will be positioned immediately adjacent to the entrance for user accessibility.

During this phase, the researchers will monitor the charging efficiency as users interface their mobile devices with the station. The primary focus is to validate the 5V DC regulation of the MT3608 boost converter and the real-time data accuracy of the Blynk dashboard in a live network environment. This setup also allows for a "User-in-the-loop" test, where the researchers can observe how the physical interaction with the mat translates into the digital telemetry displayed on the station's interface.

V. Sprint V: Evaluation and Refinement

The final sprint involves a rigorous quantitative assessment of the SmartStep system's overall performance against the initial design benchmarks. The researchers perform an End-to-End Efficiency Analysis, calculating the ratio of mechanical energy input to the final regulated DC power output. To validate the accuracy of the 'Smart' analytics, the researchers will perform a comparison between the IoT dashboard readings and a calibrated digital multimeter, targeting a margin of error below 2%.

Data collected from the faculty entrance deployment is subjected to Error Analysis, specifically comparing the State of Charge (SoC) values

reported by the IoT dashboard against high-precision multimeter readings. To enhance data integrity, the researchers implement digital signal processing techniques within the firmware, such as a Moving Average Filter or a Kalman Filter variant, to eliminate noise from the sensor readings caused by mechanical vibrations or electromagnetic interference (EMI).

The Refinement phase focuses on the thermal and mechanical optimization of the prototype. If the MT3608 boost converter exhibits significant heat dissipation under a 1A load, the researchers adjust the switching frequency or improve the airflow within the charging station housing. Furthermore, the User Interface (UI) of the mobile dashboard is refined based on feedback from the deployment phase, ensuring that the State of Health (SoH) analytics are presented via an intuitive Data Visualization layer. This sprint culminates in the final validation of the Modular Fault Tolerance—verifying that if a single PZT cell fails, the rest of the array continues to provide a charging current to the 18650 battery, resulting in a robust, deployment-ready renewable energy solution.

Respondents of the Study

The selection of respondents for this study is governed by the specific technical requirements of testing a kinetic energy harvester and its integrated monitoring system. Since the SmartStep mat must be validated against varying mechanical loads and user interactions, the following parameters are established:

Target Users:

The primary target users for the study are the faculty members, staff, students, and authorized personnel of the BSIT Department who regularly utilize the faculty entrance. This group was selected because their consistent foot traffic provides a predictable yet varied duty cycle for the piezoelectric sensors. Additionally, because the charging station is localized, these users represent the stakeholders most likely to benefit from a secondary, "green" power source for their mobile devices during operational hours.

Sample Size Distribution

To maintain a manageable yet sufficiently representative evaluation, the study uses purposive sampling to select respondents from each stakeholder category. This distribution allows the study to balance the perspectives of end-users and technical evaluators while ensuring

strong representation from IT experts, faculty and staff, and students of the SmartSteps system.

Category	**Number of Respondents**	**Percentage**
IT Experts	5	17%
Faculty and Staff	10	33%
Students	15	50%
Total Respondents	30	100%

Table 2. Respondents of the Study

Sampling Technique:

The researchers utilized purposive sampling to ensure that the participants possess the specific characteristics required to meet the study's technical objectives. IT experts were selected for their technical literacy, which is necessary to evaluate whether the State of Health (SoH) and State of Charge (SoC) algorithms function according to industry standards. Faculty and staff were purposively chosen to act as stakeholders, ensuring that the system aligns with the actual needs of

those who manage and work within the department. Finally, students were purposively chosen based on their high-traffic frequency and varying gait patterns; this variation is essential for testing the mechanical limits of the mat and observing how the TP4056 charging module responds to fluctuating energy inputs. This sampling strategy ensures that the researchers gather authentic feedback regarding the system's smart functionality, hardware resilience, and overall ease of use.

SYSTEM METHODOLOGY

4.1 External Interface Requirements

The SmartStep Platform will feature several key external interfaces to facilitate user interaction, data exchange, and system integration within the CTU Argao environment.

User Interfaces

The primary user interface will be web-based, designed to provide an interactive and responsive experience for two distinct user roles:

- Admin: The interface includes modules for managing dashboard analytics, advanced analytics, and system configuration. This encompasses firmware control, battery diagnostics, voltage threshold settings, and system restart/recalibration functions. The design prioritizes data accuracy and administrative control over the IoT network.
- User: Regular users (students and staff) have access to an interactive dashboard featuring visualized energy flow and real-time device feedback, a general analytics module displaying the sustainability scoreboard and energy harvest graphs, and a

user settings panel for profile management, system notifications, and firmware recalibration options.

Hardware Interfaces

A. Server Side: The web-based interface requires compatibility with standard desktop computers, laptops, and tablets. The physical hardware includes the Piezoelectric (PZT) sensor mat array for energy harvesting and an ESP8266 IoT controller for wireless data transmission.

B. Client Side: The system is a web-based application. Users are required to access the platform using a modern web browser (e.g., Google Chrome). The device must have an active internet connection to communicate with the IoT backend.

Software Interfaces

The SmartStep Platform will interact with IoT hardware via the ESP8266 microcontroller, which transmits real-time energy harvesting data to the backend. The frontend communicates with the backend through defined API endpoints, enabling live updates of sensor data including State of Charge (SoC), wattage output, and footfall frequency. The system

also supports remote firmware configuration through the software interface, allowing administrators to push voltage threshold settings and trigger recalibration without physical access to the hardware.

Communication Interfaces

The system's operation is highly reliant on stable network communication. It will use common internet protocols, including TCP/IP and HTTP/HTTPS, to allow access to its web-based interface and facilitate API interactions. Given the sensitivity of hardware control data, the use of secure communication protocols, especially HTTPS, is paramount. This ensures that information passed between users and the system remains confidential and unaltered. The ESP8266 Signal Strength (RSSI) is continuously monitored within the platform to verify that the IoT connection remains stable.

4.2 Functional Requirements

The functional requirements of the SmartStep Platform are structured around the interactions of two main actors: Admin and User. Both actors authenticate through a unified Login interface before being directed to their respective modules. The system encompasses general

authentication features as well as role-specific capabilities designed to promote green energy participation and provide actionable insights on the campus's piezoelectric energy harvesting system.

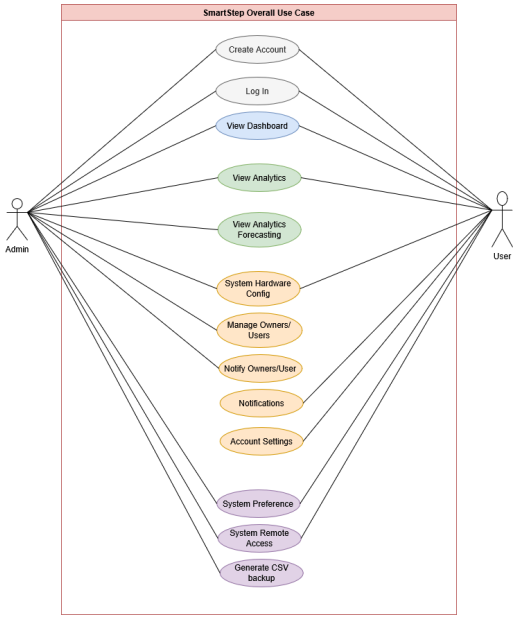

Fig 3. Whole Use Case Diagram

This is the complete use case diagram showing the interactions of two actors — Admin and User — with the entire system. Both actors authenticate via log in/log out, after which they access their respective features. The Admin manages system-level functions (dashboard analytics, advanced analytics, system configuration), while the Actor

accesses user-facing features (interactive dashboard, general analytics, user settings).

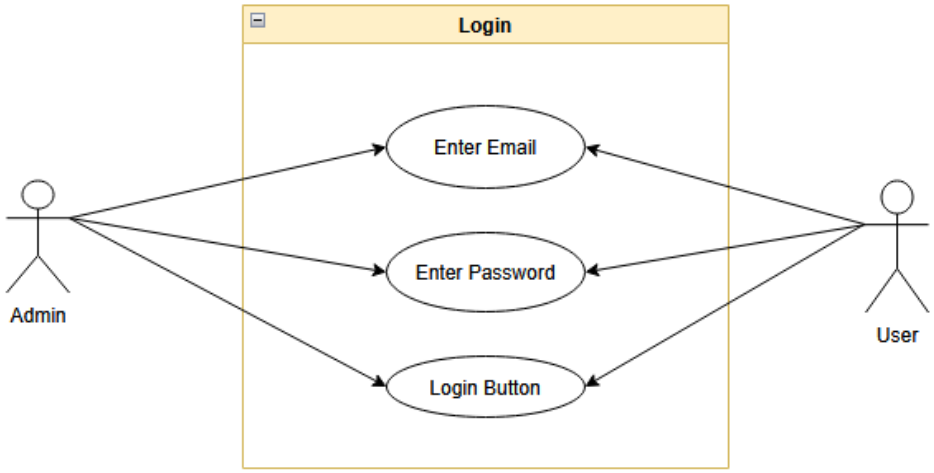

Fig 4. SmartStep Login Use Case

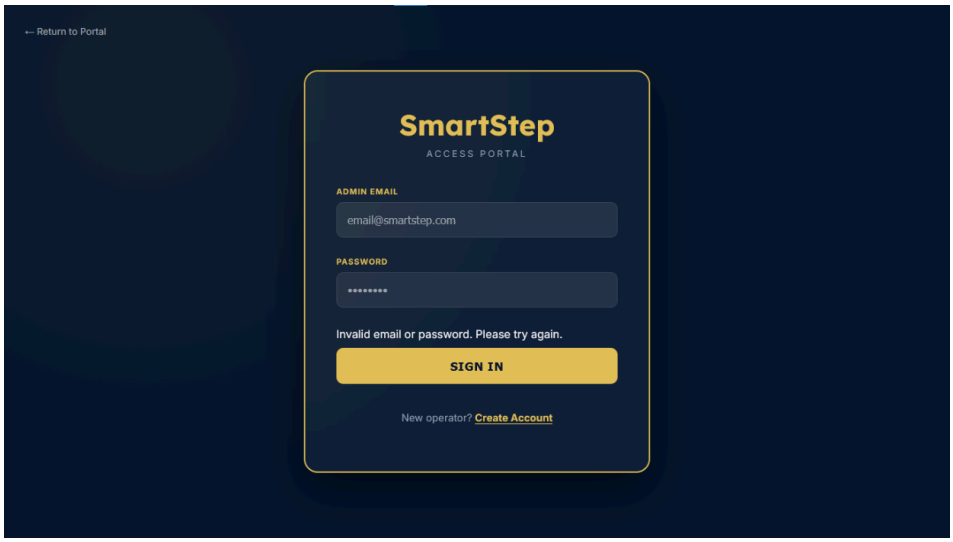

Fig 5. Log In Interface

The image above shows the Log In screen it is the primary access point for both Administrators and General Users. It is designed with a

clean, high-contrast interface that requires a registered Username and Password to enter the dashboard. This authentication layer is critical for our project because it ensures that sensitive hardware controls—like battery thresholds and system restarts—are protected from unauthorized access, maintaining the overall security of the IoT network.

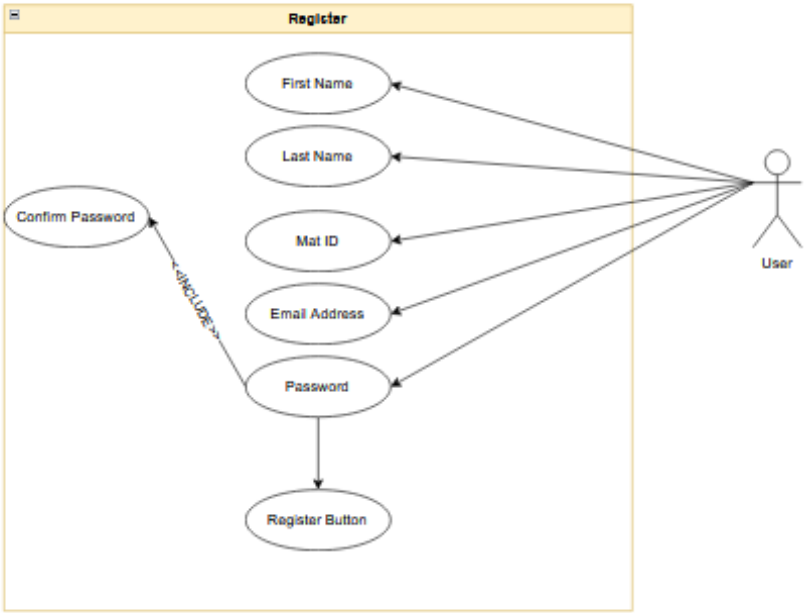

Fig 6. SmartStep Register Use Case

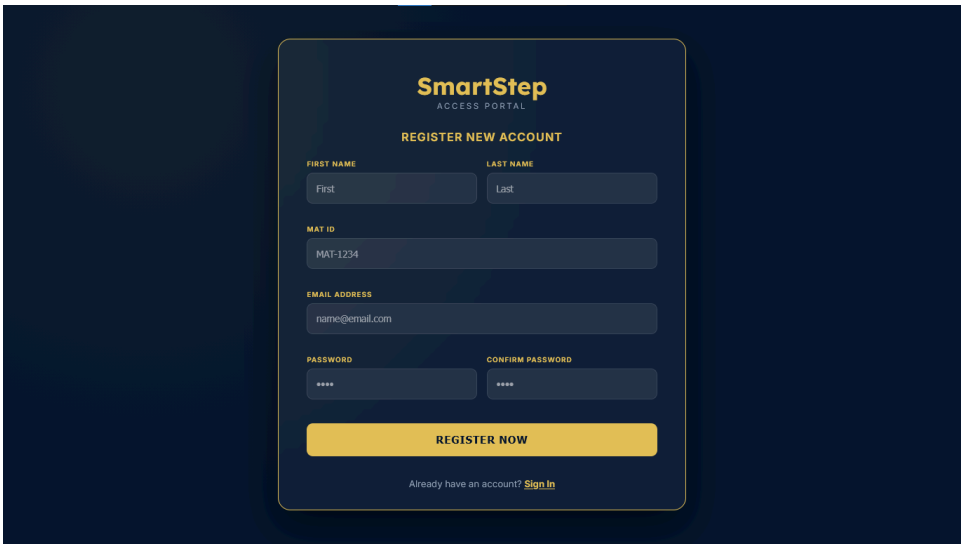

Fig 7. SmartStep Register Interface

The Register screen allows new members of the CTU faculty or the development team to create an account and join the SmartStep ecosystem. It captures essential user details, including Full Name, Email, and a unique Username. To ensure that only legitimate users can sign up, the registration process is tied to our backend database, allowing us to manage user roles and track who is interacting with the system's data and hardware settings.

Fig 8. User Use Case Diagram

This diagram focuses on the User (regular user) and their interactions after authentication. The user can access three modules: Interactive Dashboard (visual energy flow, today's status, device real-time feedback), General Analytics (sustainable scoreboard, peak usage hours, energy harvest graphs), and User Settings (profile management, system notifications, firmware control with restart/recalibration options).

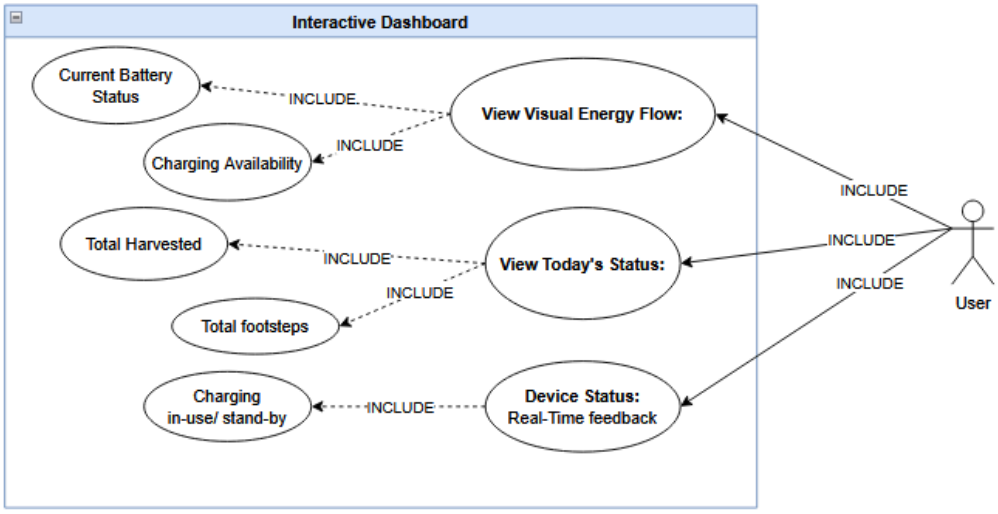

Fig 9. SmartStep User Dashboard Use Case

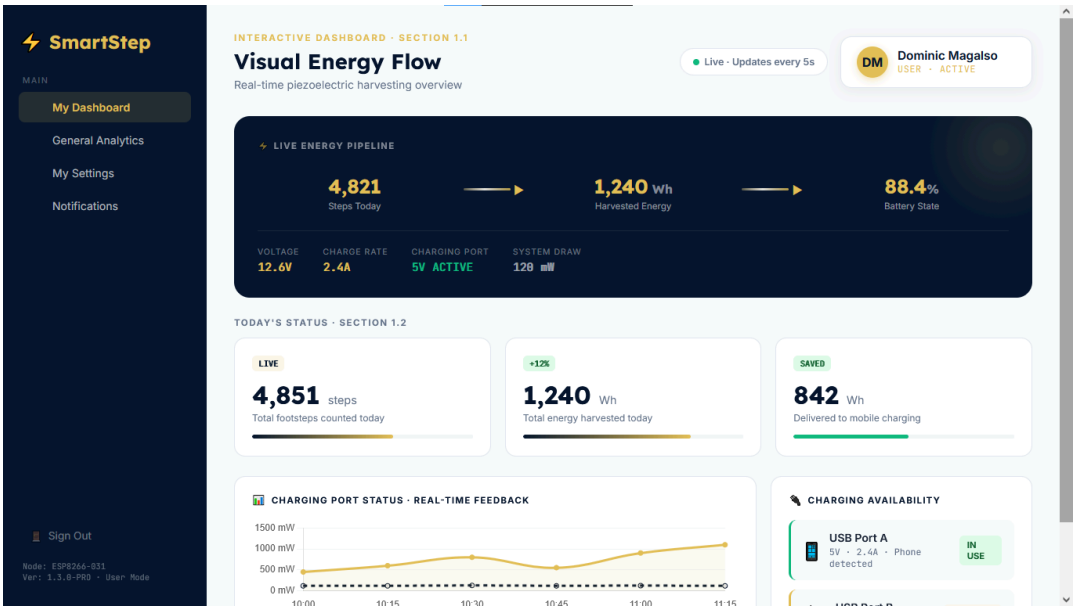

Fig 10. User Dashboard

The image above shows the User Dashboard, which serves as the primary screen for everyday interaction. It displays the live Energy Flow from the PZT mat, total Steps and Energy Harvested for the day, real-time Charging Port Status, and Battery State of Charge—giving users an instant overview of the system's availability and personal impact.

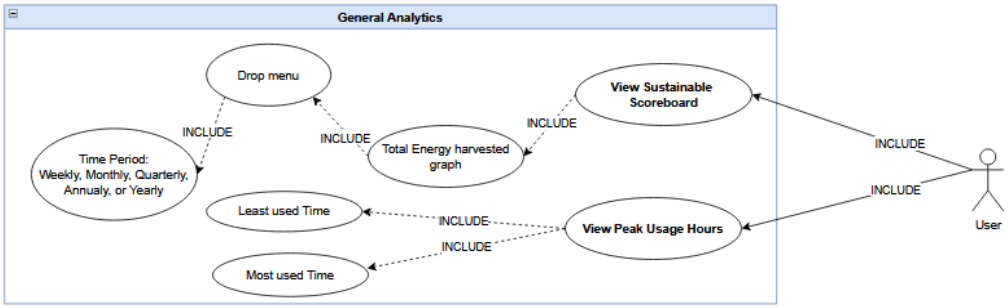

Fig 11. User Analytics Use Case

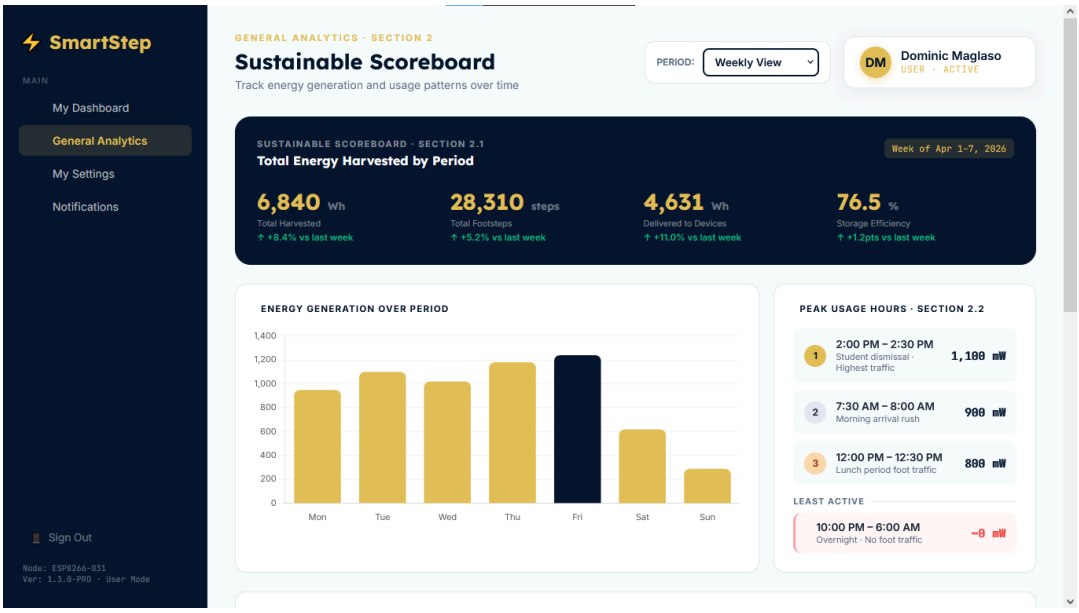

Fig 12. User Analytics

The image above shows the Analytics section, which tracks the user's Total Steps and Energy Harvested, while the Sustainability Scoreboard displays campus-wide energy generation over a selected period—transforming individual contributions into a shared environmental mission.

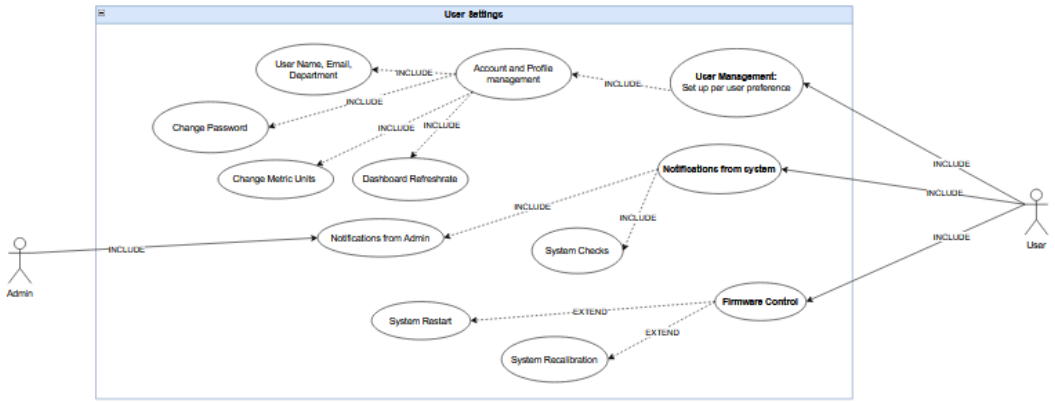

Fig 13. User Settings Use Case

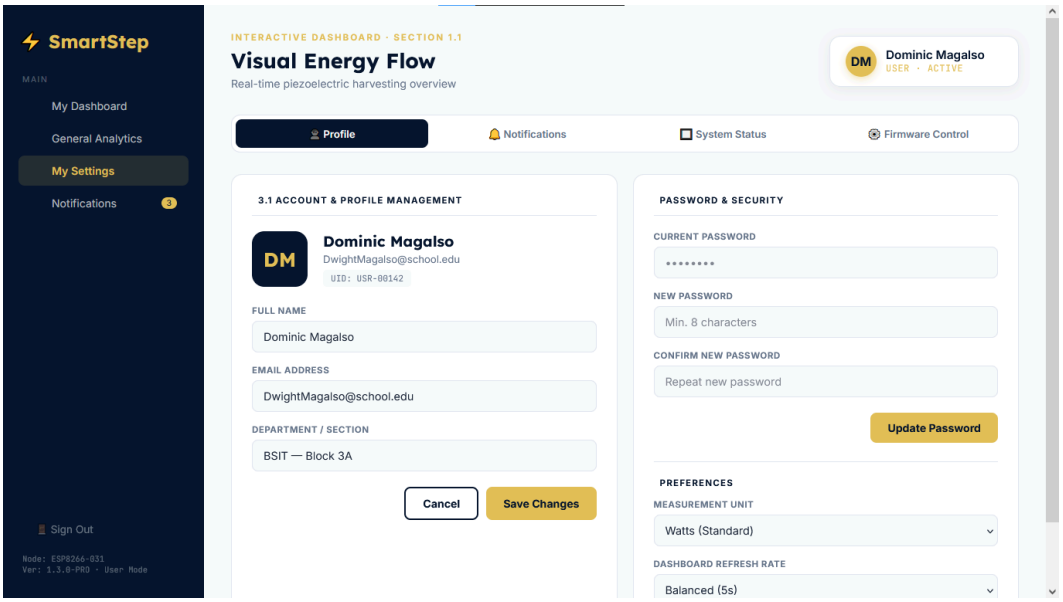

Fig 14. User Account Settings

The image above shows the User Settings module, where students and staff can manage their personal profiles, customize notification preferences, and adjust how their data is displayed on the community scoreboard—ensuring the system remains user-centric while respecting individual privacy and preferences.

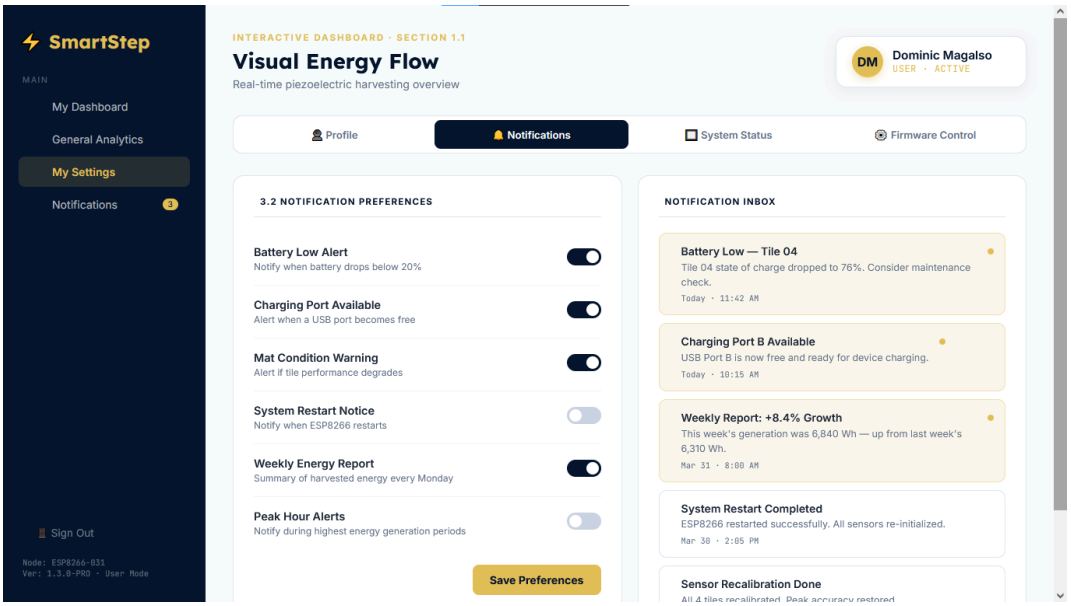

Fig 15. User Notifications

The image above shows the Notifications hub, where users receive real-time alerts such as Battery Low, Charging Port Available, and Mat Condition Warning—keeping the student body informed about the station's status and managing expectations effectively.

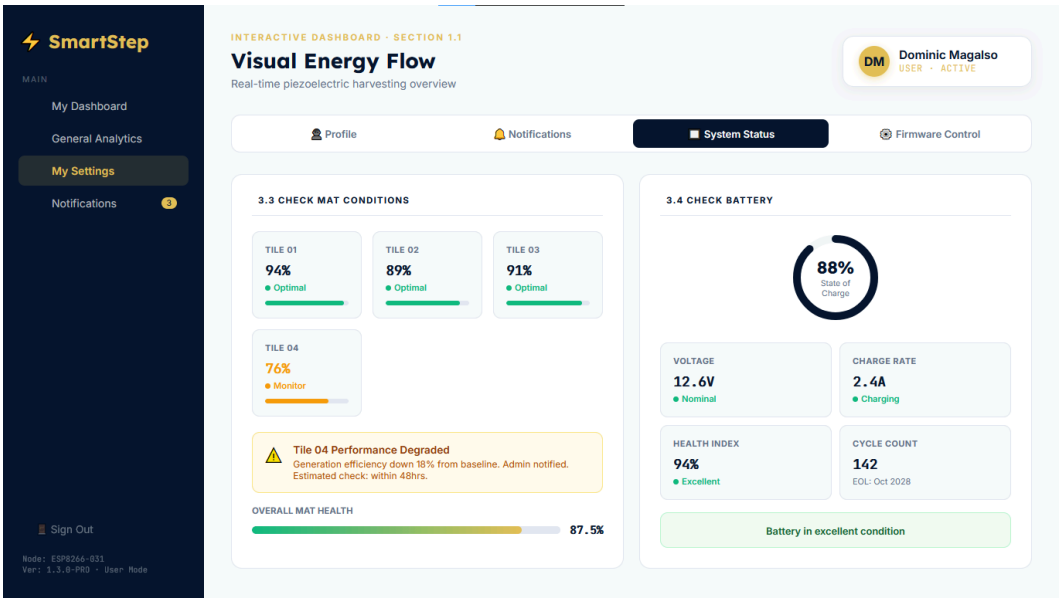

Fig 16. User System Status

The image above shows the Hardware Status section, which monitors the condition of each Mat Tile (Tiles 01–04) to catch mechanical wear early, and displays Battery Diagnostics including voltage, charge rate, health index, and cycle count—ensuring the system remains in optimal operating condition.

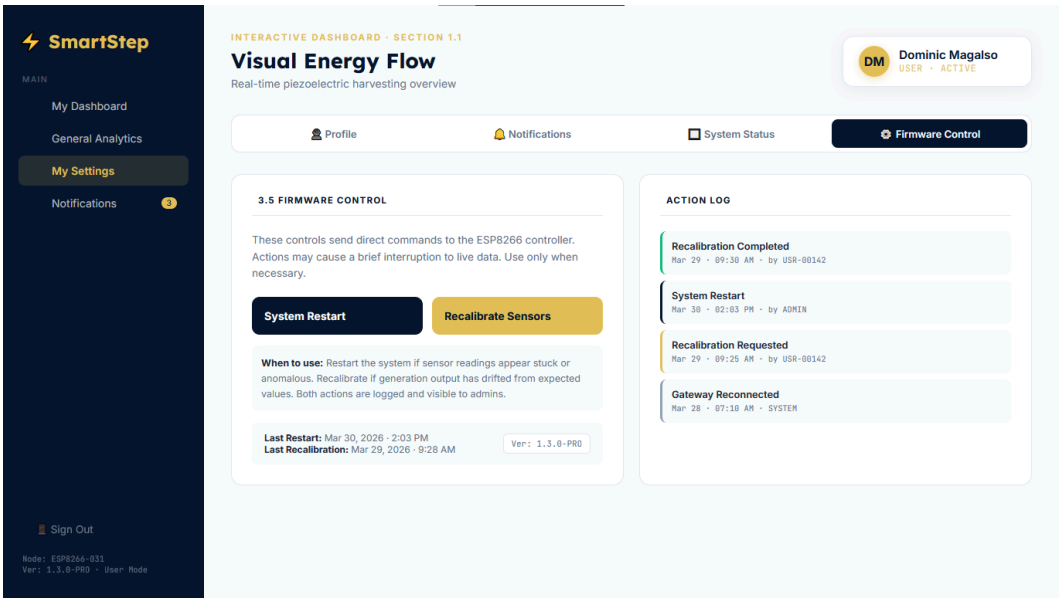

Fig 17. User Remote Firmware Config

The image above shows the Remote Config panel, where users can trigger a System Restart or Sensor Recalibration to maintain data accuracy. All actions performed are logged and visible in the Action Log, providing transparency for every command sent to the ESP8266 controller.

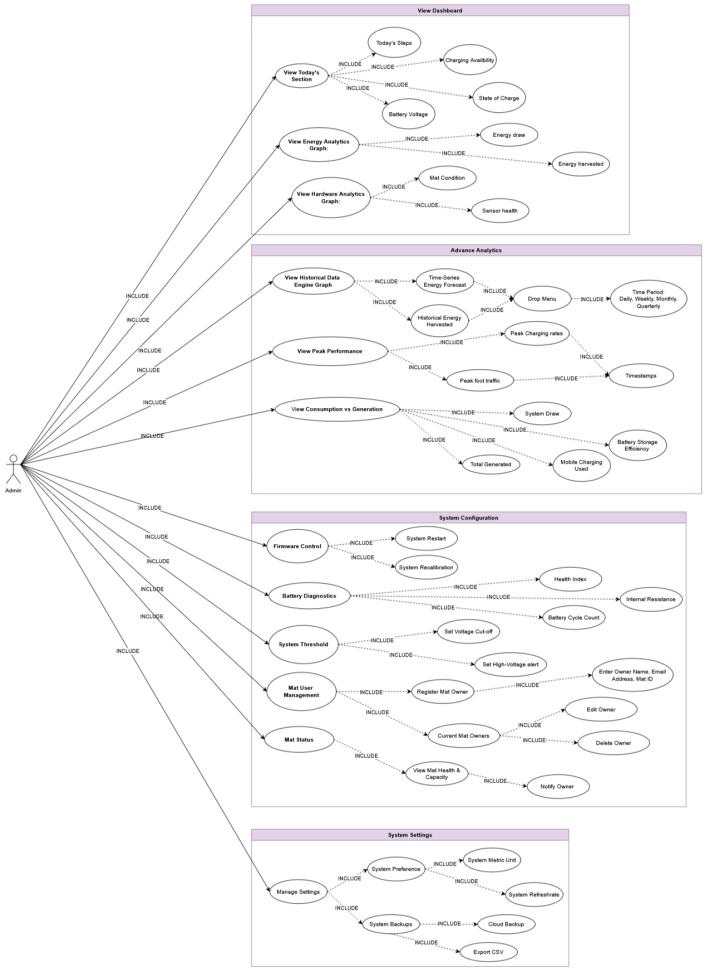

Fig 18. Admin Use Case Diagram

This diagram focuses solely on the Admin actor and their interactions with the system. After authentication, the Admin can access three main modules: View Dashboard (energy and hardware analytics), Advance Analytics (peak performance, consumption reports, historical data), and System Configuration (firmware control, battery diagnostics,

system thresholds). The Admin can also extend to manage system settings like preferences and backups.

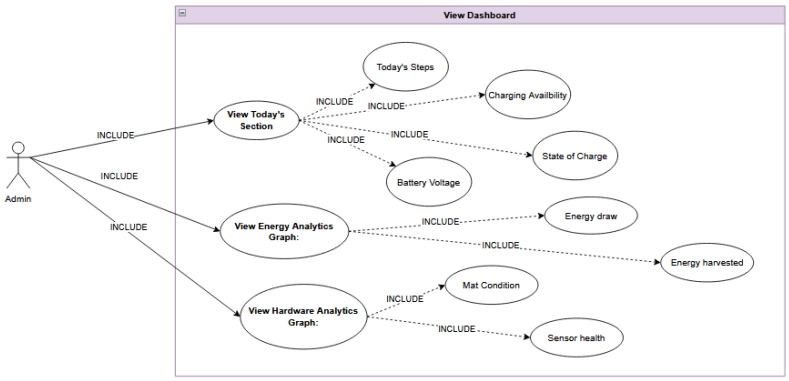

Fig 19. Admin Dashboard Use Case

Fig 20. Admin Dashboard

The image above shows the Admin Dashboard, which provides a high-level overview of the system's vitals, including live Energy Flow from the PZT mat, State of Charge (SoC), real-time Wattage, and Sensor Health—allowing administrators to monitor power generation and battery health at a single glance.

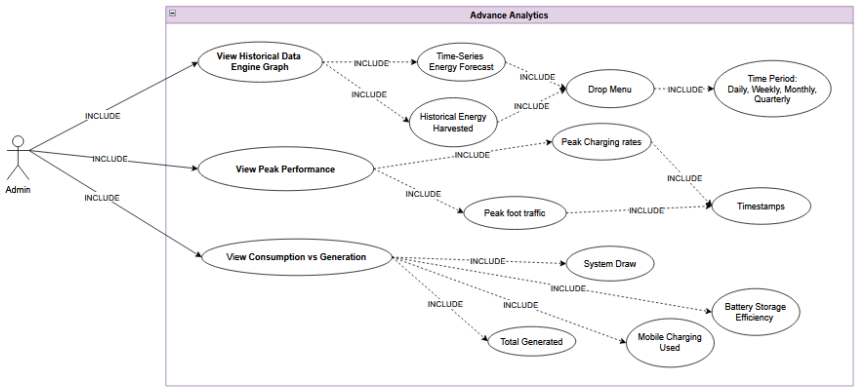

Fig 21. Admin Analytics Use Case

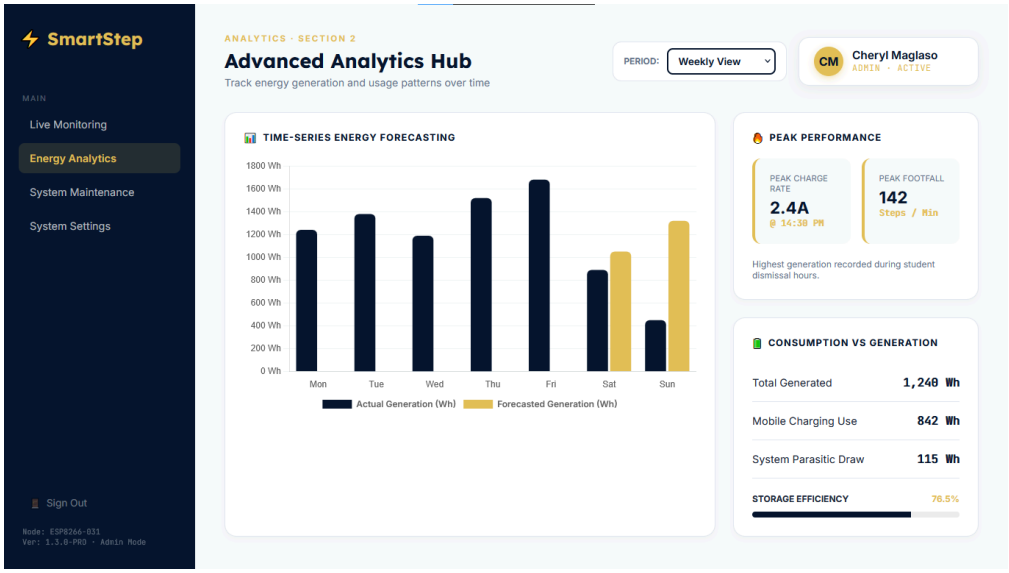

Fig 22. Admin Analytics

The image above shows the Advanced Analytics Hub, which tracks long-term performance trends through Energy Harvested vs. Parasitic Draw, Footfall Frequency, and Consumption vs. Generation breakdown—allowing administrators to evaluate power conversion efficiency and identify peak energy generation periods on campus.

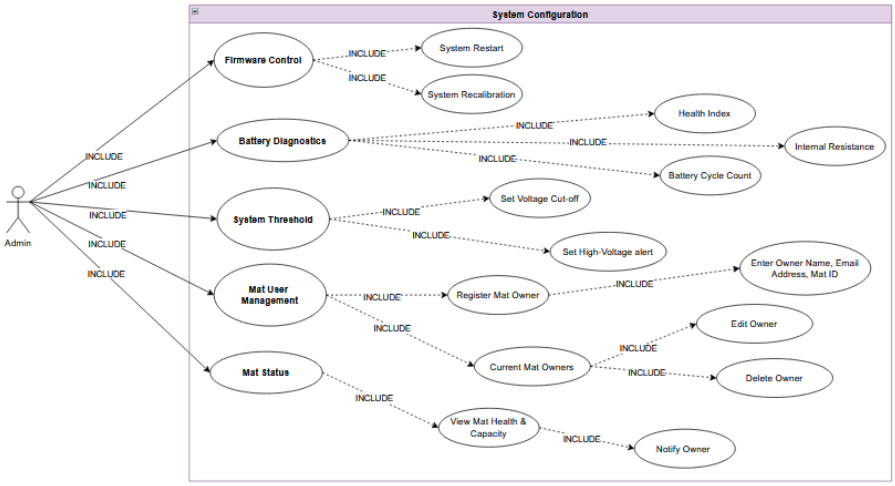

Fig 23. Admin Hardware Control

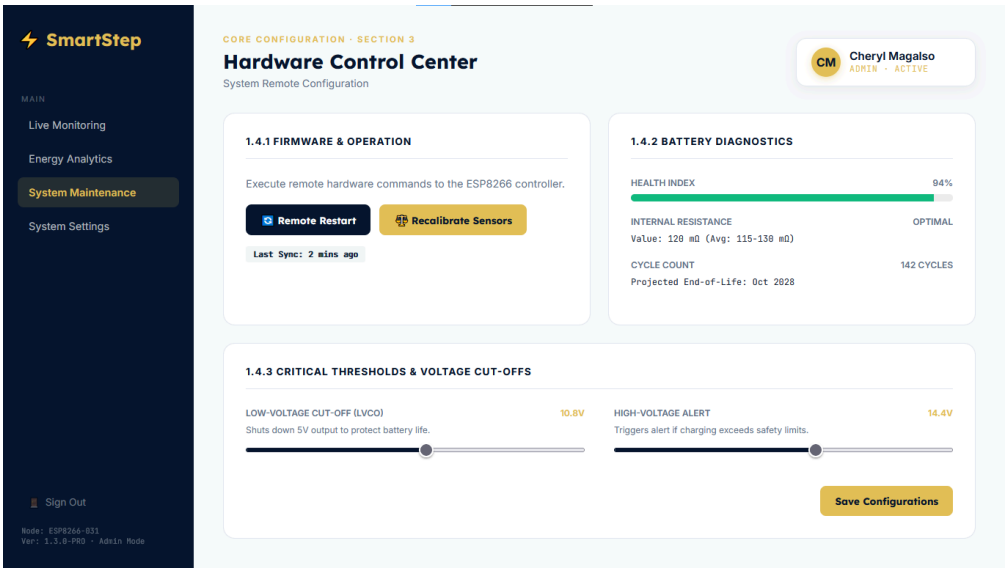

Fig 24. Admin Hardware Control

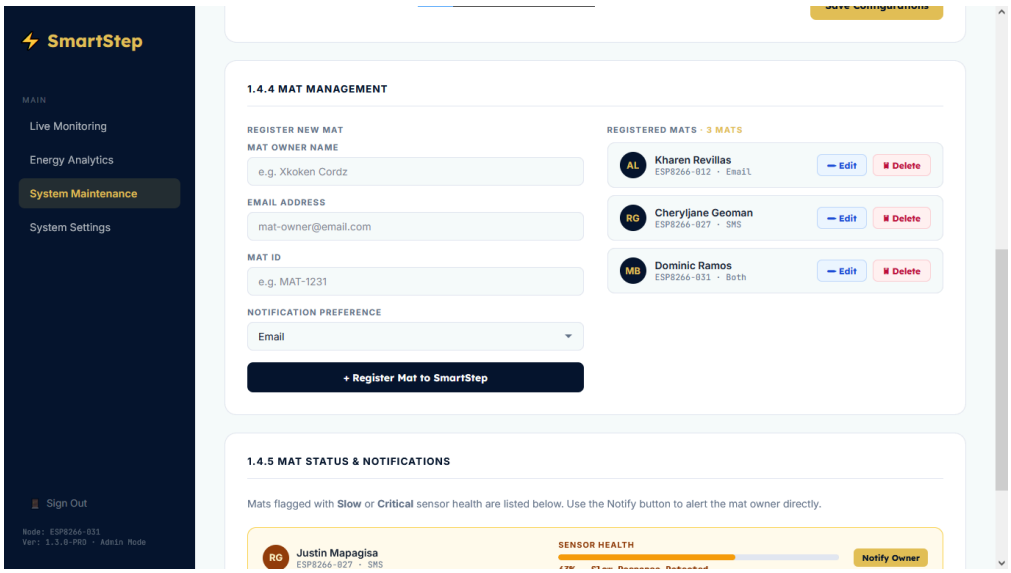

Fig 25. Admin Hardware Control

The image above shows the Hardware Control Center, where administrators can trigger System Restarts or Sensor Recalibrations, set Low-Voltage Cut-Off and High-Voltage Alert thresholds, and monitor Battery Diagnostics—ensuring the system remains accurate and protected from damage.

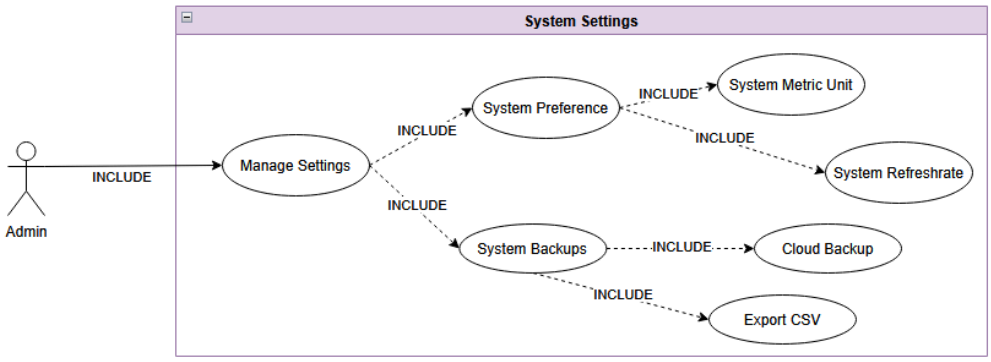

Fig 26. Admin System Settings

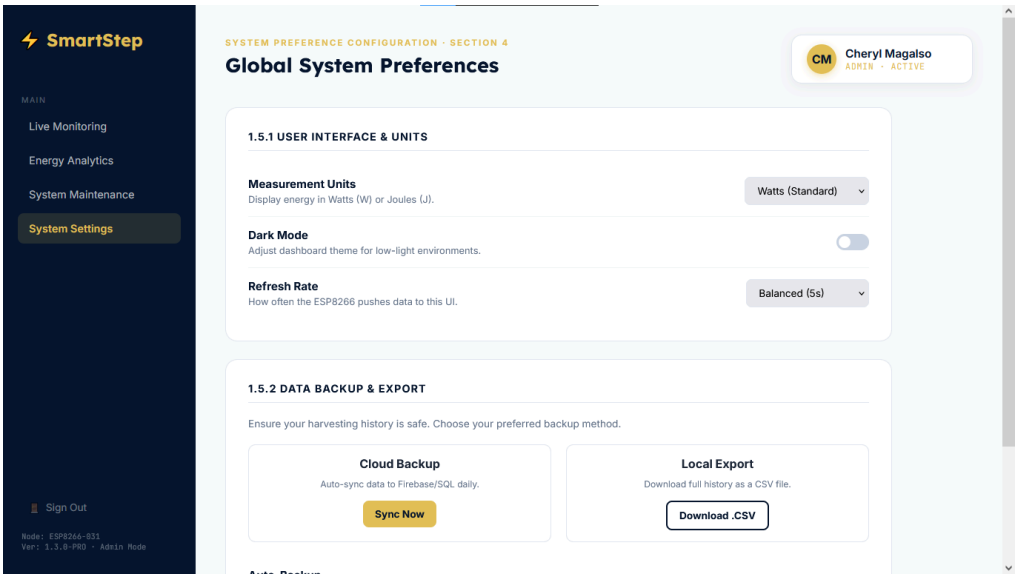

Fig 27. Admin System Settings

The image above shows the Settings Page, which allows administrators to update their Admin Profile, manage security credentials, and configure system-wide preferences such as measurement units, dark mode, dashboard refresh rate, and data backup options—ensuring authorized faculty can effectively manage the system.

4.4 Data Design

The data design for the system follows a sequential flow that translates physical movement into actionable digital insights. It begins with the acquisition of analog signals from the piezoelectric transducers, which are then processed by the ESP8266 to calculate voltage and estimated wattage. This data is packaged into a structured format and transmitted via an IoT protocol to a centralized database. The design ensures data integrity by implementing validation checks at the server level, ensuring that only accurate energy metrics are stored for historical analysis and real-time visualization on the user and admin dashboards.

4.5 Class Diagram

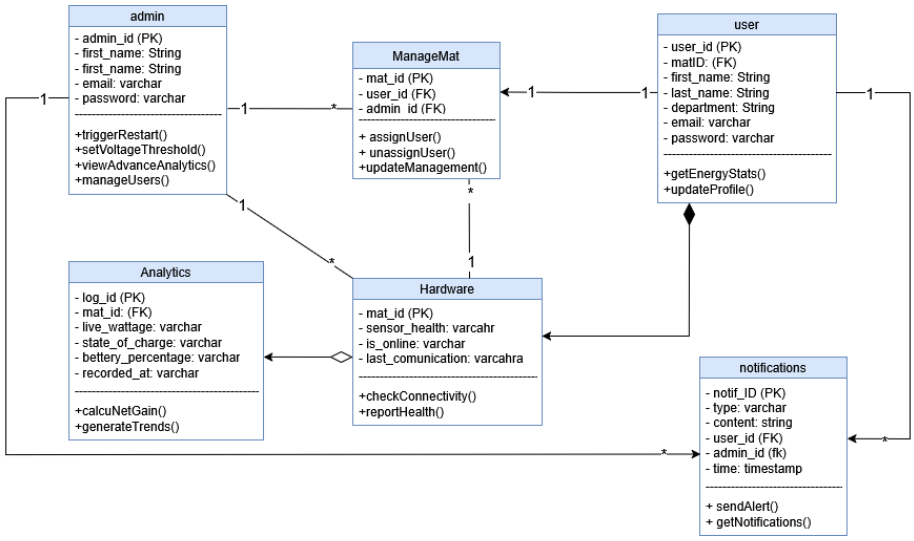

Fig 28. SmartStep Class Diagram

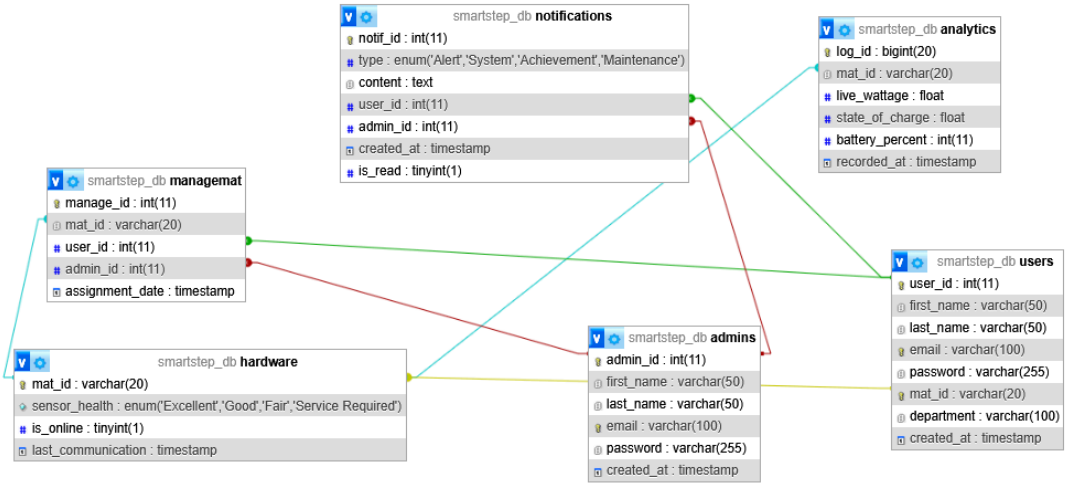

Fig 29. SmartStep Database Schema

The class diagram outlines the object-oriented structure of the web application, defining the relationships between the various entities within the system. It centers on a base user class that branches into specific "Admin" and "Users" roles, each with unique permissions and methods.

Supporting classes manage the hardware interactions, such as battery health diagnostics and sensor calibration logic. By defining these classes, the system can efficiently handle complex tasks like updating the sustainability scoreboard or triggering system restarts, ensuring the code remains modular, scalable, and easy to maintain.

4.6 System Architecture

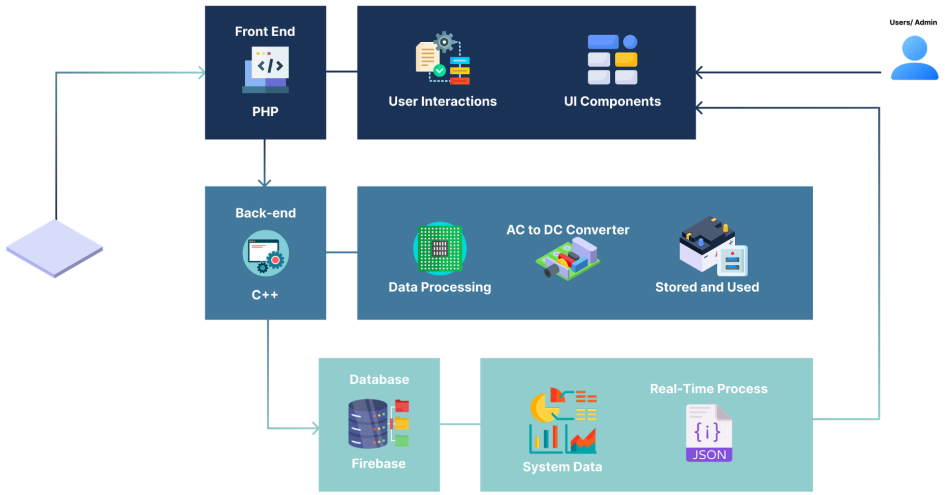

Fig 30. SmartStep System Architecture

The SmartStep system architecture is a multi-layered IoT framework designed to bridge physical kinetic energy harvesting with

real-time web analytics. It is structured into distinct functional layers: the physical perception layer (the PZT mat and bridge rectifier circuit), the power/storage layer (the 12V Dyna Power battery and buck converters), the communication layer (the NodeMCU utilizing campus Wi-Fi), and the application/service layer (the PHP/MySQL backend and frontend dashboards). This architectural design prioritizes high availability and security, utilizing an API Gateway and load balancers to manage incoming data traffic. By segregating business logic into distinct microservices—specifically separating administrative controls from user-facing data presentation—the architecture ensures that critical hardware management functions are isolated, responsive, and secure from unauthorized access.

4.7 Security and Data Privacy Requirements

To ensure the integrity and confidentiality of student data, the system implements a multi-layered security protocol centered on the Firebase ecosystem and is designed in strict compliance with the Data Privacy Act of 2012 (RA 10173). Authentication is handled via Firebase Auth, requiring institutional credentials for access, while Authorization is enforced through granular Firebase Security Rules that restrict users to viewing only their assigned mat_id data. Adhering to the principle of data

minimization, the system only collects essential identifiers. Furthermore, all telemetry data and personal information are protected by SSL/TLS encryption during transit between the ESP8266 edge device and the cloud, ensuring that energy harvesting logs remain tamper-proof and private.

4.8 Other Non-Functional Requirements

Beyond core functionality, the system must satisfy several non-functional pillars to ensure long-term viability. Usability is achieved through a mobile-first, responsive dashboard design that simplifies complex electrical data into intuitive gauges for the end-user. Scalability is a core feature of the NoSQL data design, allowing the system to integrate additional PZT mats without requiring a restructuring of the database. For Maintainability, the software architecture is modular, separating sensor logic from cloud synchronization code to facilitate easy hardware troubleshooting. Finally, Portability ensures that the web-based presentation layer is cross-browser compatible, providing a consistent user experience across Android, iOS, and various desktop operating systems.

4.9 Software Requirements

Programming languages

The SmartStep system utilizes a dual-language approach to manage the high-frequency requirements of IoT data processing. C++ (Arduino Core) is the primary language for the NodeMCU ESP8266 firmware, chosen for its efficiency in handling low-level hardware interrupts from the PZT sensors and managing precise voltage calculations for the 12V battery diagnostics. Complementing the hardware logic, JavaScript (ES6+) is used for the web-based presentation layer to handle asynchronous data streams from the cloud, ensuring that the user interface remains reactive and updates without manual page refreshes.

Frameworks

To ensure a robust and responsive user experience, the project leverages the Bootstrap 5 frontend framework, providing a mobile-first grid system and pre-styled components that adapt to various screen sizes. For the backend, the Firebase SDK acts as the core framework for real-time synchronization. By utilizing the Firebase Realtime Database and Authentication libraries, the system eliminates the need for traditional server-side frameworks (like Laravel or Express), allowing for a streamlined, serverless

architecture that pushes data directly from the ESP8266 to the end-user.

Tools

Development and debugging are facilitated through a suite of specialized software tools. Visual Studio Code (VS Code) and the Arduino IDE serve as the primary Integrated Development Environments (IDEs) for authoring the web frontend and hardware firmware, respectively. Postman is employed to test API connectivity and JSON data structures during the integration phase. Additionally, the Firebase Console is used as an administrative management tool to monitor database traffic, manage user credentials, and review historical telemetry logs for performance validation.

4.10 Operational Requirements and Cost Considerations

Category	Item	Estimated Cost (PHP)
Microcontroller	NodeMCU ESP8266	₱250.00
Power Storage	12V Dyna Power Battery	₱1,500.00
Sensors	PZT Piezoelectric Array (Pack)	₱500.00
Monitoring	INA219 Current Sensor	₱180.00
Charging Station	DC-DC Buck Converter (QC 3.0)	₱220.00
TOTAL		**₱2,650.00**

Fig 31. SmartStep Estimated Operational Cost

The success of SmartStep depends on a reliable Wi-Fi connection and careful power management. The system will require a steady 2.4GHz Campus Wi-Fi signal for the ESP8266 to send data to Firebase. To keep the 12V Dyna Power battery healthy, the system is designed to monitor voltage levels and send alerts if the battery needs attention. Financially, the project aims to be low-cost by using the Firebase Free Tier, which eliminates monthly hosting fees. With a total estimated hardware cost of ₱2,650.00, this project is intended to show that a sustainable energy tracker can be built affordably, making it a feasible solution for future use across the campus.

4.11 Schedule Feasibility

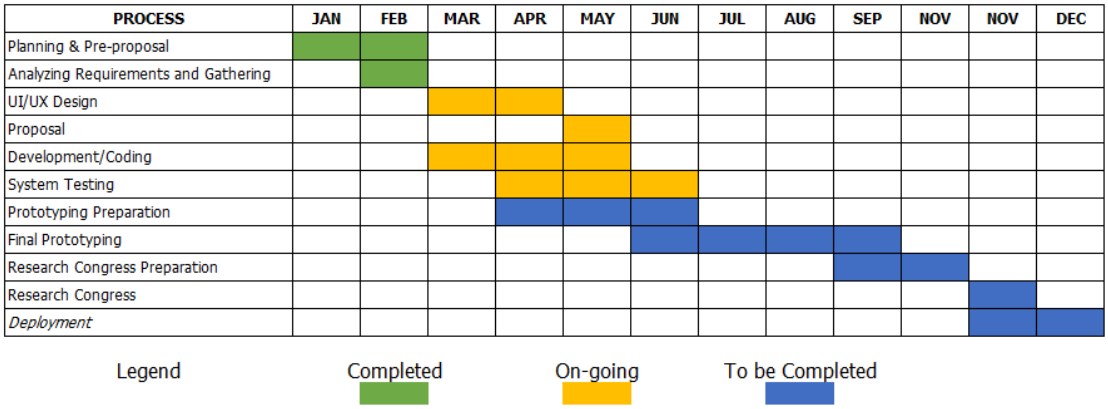

Fig 32. SmartStep Schedule Feasibility Gantt Chart

The project schedule is organized into seven core phases following a hybrid Iterative-Waterfall SDLC model. This structure allows the team to maintain a linear progression for administrative milestones (such as Title Hearing and Research Congress) while utilizing an iterative "Build-Test" loop during the middle of the year. By overlapping the Continuous Testing phase with Iterative Development, the project ensures that the high-frequency data generated by the PZT sensors is correctly synchronized with the Firebase Realtime Database before the final physical assembly. This phased approach guarantees that the 12V charging station and the administrative analytics dashboard are fully validated through multiple stress-test cycles prior to deployment.

DEFINITION OF TERMS

To ensure clarity and consistency throughout this study, the following key terms are defined:

18650 Lithium-Ion Battery – A standardized rechargeable battery cell that serves as the central energy reservoir for storing harvested power and providing a stable source for the charging station (Song et al., 2019).

Bridge Rectifier (DB107/1N4007) – A circuit component used to convert the alternating current (AC) spikes generated by piezoelectric discs into direct current (DC) suitable for battery storage (Covaci & Gontean, 2020).

Deep Sleep Mode – A power-saving logic programmed into the ESP8266 microcontroller that reduces standby energy consumption during periods of inactivity to ensure that the system's parasitic load does not exceed the energy harvested (Radfar et al., 2020).

Direct Piezoelectric Effect – The process of converting mechanical stress or pressure from human footsteps into electrical power through the use of piezoelectric materials (Covaci & Gontean, 2020).

Intelligent Battery Management System (BMS) – A digital framework that provides active monitoring and a digitalized exchange of data to ensure

harvested energy is managed through optimized charging cycles and safety cut-offs (Tharun et al., 2021).

IoT Gateway (ESP8266) – A Wi-Fi-enabled microcontroller used for event-driven data logging and transmitting system analytics, such as voltage and current, to a mobile dashboard via the I2C protocol (Selim et al., 2024).

MT3608 Boost Converter – A DC-to-DC power module that increases the fluctuating voltage from the lithium-ion battery to a steady 5V output required for standard USB charging interfaces (Song et al., 2019).

State of Charge (SoC) – The measurement of the current energy level in the battery relative to its capacity, utilized by the system to prevent overcharging or deep discharge (Texas Instruments, 2015).

State of Health (SoH) – A metric used to estimate the remaining useful life of the battery storage unit based on its charging cycles and internal resistance over time.

TP4056 Charging Module – A hardware component that safely manages the storage of harvested energy into the 18650 battery by regulating the constant-current/constant-voltage (CC/CV) flow between the harvester and the storage (Xie & Zhang, 2021).

Zener Diode – A semiconductor device used in this project as a voltage clamp to protect the sensitive inputs of the ESP8266 and INA219 from high-voltage transients generated by the piezoelectric array.

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